NexaDS: Automated Data Science & Predictive Analytics

Comprehensive Experiment & Machine Learning Audit Report

Experiment ID:	exp_c4ca38b64a	**Target Feature:**	churn
Problem Type:	Classification	**Best Selected Model:**	K-Nearest Neighbors
Generated Timestamp:	2026-11-12 13:25:14	**Audit Status:**	Validated & Complete

1. Dataset Information

The analysis was conducted on dataset **Customer_Churn_Classification.csv**, comprising **250 rows** and **7 columns**.

2. Dataset Quality Diagnosis

Computed Dataset Integrity Score: **100.0 / 100**

3. Data Preprocessing Pipeline

Operations Applied: Standardization
Rows before: 250 | Rows after: 250 | Missing items handled: 0

4. Baseline Model Training & Benchmarks

Algorithm	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.6	0.5269	0.6	0.543
Decision Tree Classifier	0.58	0.4767	0.58	0.5096
Random Forest Classifier	0.58	0.5093	0.58	0.5296
Support Vector Machine	0.66	0.4356	0.66	0.5248
K-Nearest Neighbors	0.66	0.6389	0.66	0.6427
Gradient Boosting Classifier	0.64	0.6018	0.64	0.6041

5. Model Error & Weakness Analysis

- **Weakness Identified:** Class 'Yes' has low recall (35.3%), missing 11 actual instances.
- **Weakness Identified:** Class 'Yes' suffers from high false alarms with 6 false positive predictions.

6. Retraining & Hyperparameter Tuning (Before vs After)

Tuning Techniques: Cross-validated GridSearch, Hyperparameter Optimization

Metric	Baseline	Improved	Difference
Accuracy	0.66	0.58	-0.08

Precision	0.6389	0.598	-0.0409
Recall	0.66	0.58	-0.08
F1 Score	0.6427	0.587	-0.0557

7. Explainable AI (XAI) & Feature Ranking

Top 5 Influential Predictors: **account_length, monthly_charges, total_charges, contract_type, internet_service**
• Permutation Feature Importance validated the dominant predictive weights.
• Non-linear boundaries and feature interaction accounted for variance distribution.

8. Final Engineering Observations

The automated pipeline identified **K-Nearest Neighbors** as the top performing architecture for predicting **churn**. The model and its preprocessing column transformer have been serialized and persisted for dynamic prediction inference. All metrics and audit artifacts adhere to reproducible data science standards.